Water hydraulics project brief

Prepare an engineering discussion.

This worksheet is not a component selection, approved design or quotation.

Include metric and US customary units where applicable (US gallons for flow). Mark unavailable values “Confirm”.

1. APPLICATION

Organisation / project:

Contact:

What must the equipment do?

Why is avoiding hydraulic oil important here?

Existing equipment or new installation?

Consequence of a leak or loss of motion:

2. LOAD AND MOTION

Required push force (kN or lbf):

Required pull force (kN or lbf):

Travel / stroke (mm or in):

Target extension time (seconds):

Target retraction time (seconds):

For rotary motion: torque (N.m or lbf.ft) and speed (rpm):

Load orientation, external forces and changing loads:

Cycles per hour and operating hours per day:

Maximum holding time and acceptable drift:

Emergency or fail-safe position:

3. FLUID AND ENVIRONMENT

Proposed water source (tap / treated / seawater / other):

Available water analysis and sampling date:

Chlorides / salinity, pH, hardness and conductivity, with units:

Particles / cleanliness and existing filtration details:

Normal / minimum / maximum fluid temperature:

Ambient temperature range during operation and shutdown:

Freeze exposure and power-loss conditions:

External washdown, chemicals, salinity or immersion:

Treatment, biological control and stagnation conditions:

Water supply pressure / flow and drainage limitations:

4. CIRCUIT AND COMPONENTS

Required operating pressure and flow (include units):

Existing pump, valve, actuator, seal and hose details:

Confirmed permissible fluid and component ratings:

Pipe / hose lengths, diameters, elevations and routing:

Pressure protection, load holding and isolation requirements:

Power supply and control interfaces:

Space, access, noise and service constraints:

Relevant requirements, acceptance criteria and test procedure:

5. EVIDENCE AND NEXT STEP

Circuit schematic, drawings and photographs available:

Load calculations and duty profile available:

Catalogue data / test data / calculated assumptions (identify each):

Unknowns to confirm:

Target timing and budget constraints:

Requested next step (feasibility / component review / pilot / quote):

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Calculators give ideal illustrative values. Final design requires component-specific ratings, losses, operating limits and safety review.